

Tables and Figures for the MangoLeafVarietyBD Dataset

This document contains all tables and figures used in the manuscript.

Table 1: Comparison of MangoLeafVarietyBD with existing Bangladeshi datasets.

Mango Dataset	Number of Classes	Total Raw Images	Resolution	Geographical Coverage
BDMANGO [2]	6	837	640 x 480 px	Dhaka, Bangladesh
MangoLeafVarietyBD	8	2,744	Up to 4000 x 3000 px	Thakurgaon & Savar, Bangladesh

Table 2: A quick look at the mango leaf variety dataset.

Data type	Raw field-captured mango leaf images (original resolution; no resizing applied in dataset)
Data Format	Images are JPG.
Total images	2,744 raw images.
Number of classes/varieties	Total of eight classes (Amrupali, Banana, Bandigori, Himsagar, Surjapuri, Harivanga, Kacha Mitha, and Brunei King).
Distribution of original images	Amrupali: 308 images, Banana: 351 images, Bandigori: 332 images, Himsagar: 341 images, Surjapuri: 370 images, Harivanga: 366 images, Kacha Mitha: 376 images, and Brunei King: 300 images.
Acquisition method	Snapping of the samples was done on white background using two mobile phones after sample collection and the two sides of a leaf were captured to have an in-depth picture.
Data source locations	Thakurgaon District and Daffodil International University, Savar, Dhaka.

Where applicable	The dataset can be used in classifying the mango leaf varieties, image classification, phenotyping of plants, and agricultural studies that use machine learning.
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Table 3: Architecture-Specific Hyperparameters Used for Model Training

Model	Learning Rate	Dropout	Weight Decay	Label Smoothing
ViT-B16	2×10^{-4}	0.8	0.50	0.50
DenseNet121	1.4×10^{-3}	0.8	0.50	0.50
GhostNetV2	5×10^{-5}	0.8	0.30	0.40
ResNet18	1×10^{-4}	0.0	0.00	0.00

Table 4: Benchmark Performance of Deep Learning Models on the MangoLeafVarietyBD Test Set (N = 1,921)

Model	Test Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)	Latency (ms)	Throughput (FPS)
ViT-B16	0.9755	0.9762	0.9755	0.9756	13.6739	~73.13
DenseNet121	0.9558	0.9565	0.9558	0.9558	3.5861	~278.85
GhostNetV2	0.9474	0.9482	0.9474	0.9474	1.5566	~642.42
ResNet18	0.9391	0.9417	0.9391	0.9394	1.4083	~710.07



Fig. 1. Examples of eight distinct types of original mango leaf images included in the dataset.

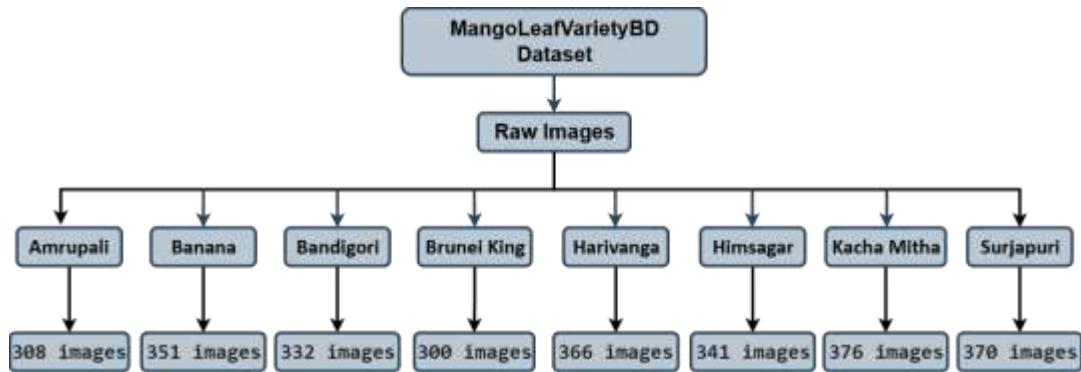


Fig. 2. Structure of the directory of the dataset.



Fig. 3. Flowchart showing the data preparation steps.

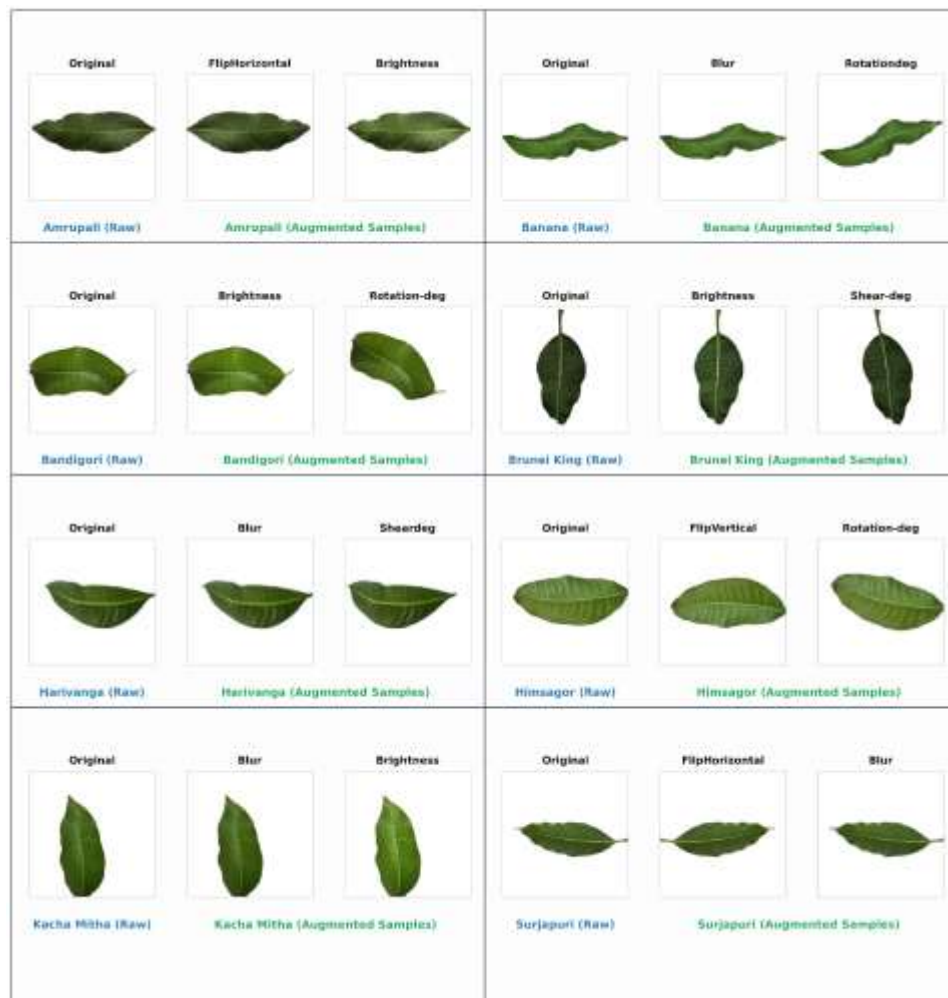
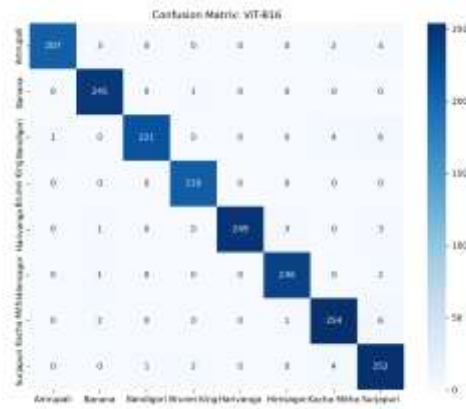
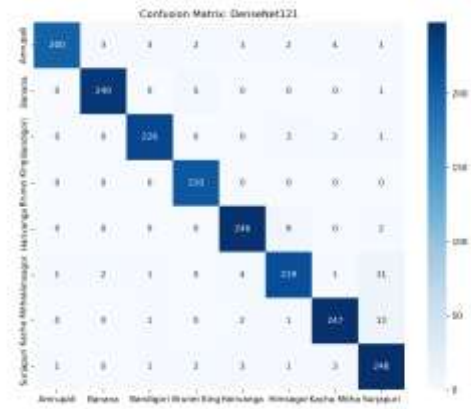


Fig. 4. Sample of augmented images of MangoLeafVarietyBD Dataset (for Experimental Model Training).



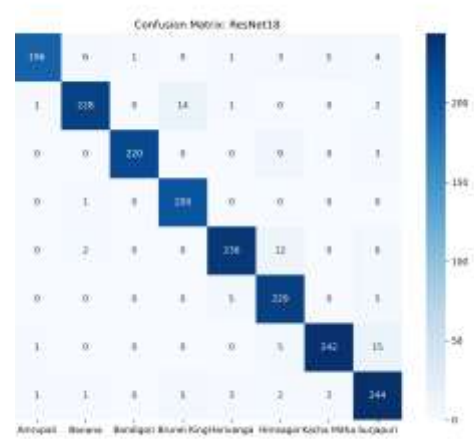
(a)



(b)

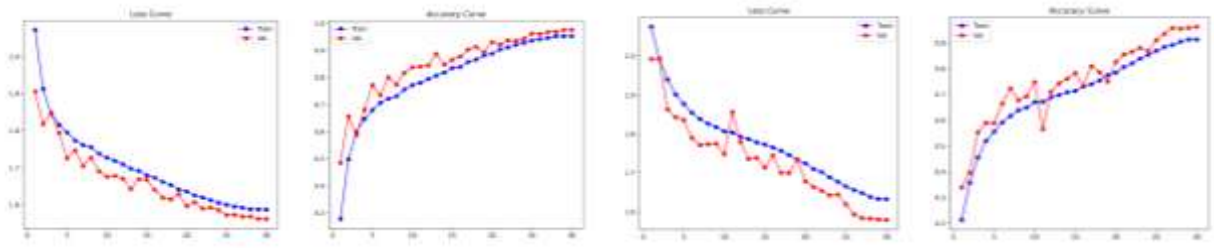


(c)



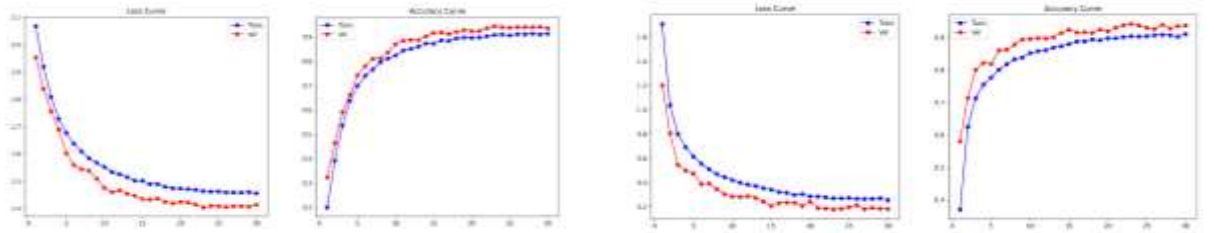
(d)

Fig. 5. Confusion Matrices of Four Deep Learning Models on MangoLeafVarietyBD Test Set (N=1,921).
(a) ViT-B16, (b) DenseNet121, (c) GhostNetV2, (d) ResNet18



(a)

(b)



(c)

(d)

Fig. 6. Training Dynamics (Loss and Accuracy Curves) of Four Deep Learning Models on MangoleafVarietyBD. (a) ViT-B16, (b) DenseNet121, (c) GhostNetV2, (d) ResNet18